Course: BTech **Semester:** 2

Prerequisite: strong foundation in programming and object oriented concepts.

Rationale : OOP in C++ provides a structured approach to programming by enabling encapsulation, inheritance, polymorphism, and abstraction. It enhances code reusability, maintainability, security, and scalability, making it a preferred choice for building large-scale software applications

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
2	-	2	-	3	20	20	20	60	30	150

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Principles of Object Oriented Programming**Object Oriented Languages, Applications of Object Oriented Programming, C++ Concepts,Structure of C++ program,Basic Data types in C++, User defined Data types, Derived Data types,Declaration of variables and Dynamic initialization of variables, Reference variables,Operators in C++, Scope Resolution Operators,Differentiate OOP and POP, Access Specifiers: Public, Private, Protected	25	10
2	**Functions, Class and Objects** Concept of Functions in C++, Function prototyping, Function Defining, Function Calling, Inline Function, Default Arguments, Constant Arguments, Function Overloading,Defining Class and Creating Objects,Defining Member function , Nesting of Member functions, Private Member functions, Friend function	25	10
3	**Constructor and Destructor** Constructor, Types of Constructor: Default Constructor, Parameterized Constructor, Copy Constructor, Destructor	20	8
4	**Inheritance** Inheritance , Defining Derived Classes, Types of Inheritance : Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hybrid Inheritance, Function Overriding,Function Overloading vs. Function Overriding,Virtual Base Class , Abstract Classes.	20	8
5	**Pointer & Virtual Functions** Pointers to objects,Pointer to Derived Classes , Virtual Functions, Pointer to Virtual Functions, 'this' Pointer.	10	6
	Total	**100**	**42**

Reference Books

1.	**Object Oriented programming with C++ (TextBook)** By E. Balagurusamy | TMH
2.	**The Complete Reference C++, (TextBook)** By by Herbert Schildt | TMH
3.	**Object Oriented Programming with ANSI and Turbo C++** By Ashok Kamthane | Pearson

Course Outcome

After Learning the Course the students shall be able to:
After Learning the Course the students shall be able to: 1. Perform various Program of functions in C++. 2. Perform class and object concept of program. 3. Apply constructors and destructors for object initialization and cleanup. 4. Demonstrate various inheritance types and function overriding. 5. Utilize pointers and virtual functions for dynamic behaviors.

List of Practical

1.	**Write a program to Display Hello World on your Output Screen in C++.**
2.	**Write a C++ program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.**
3.	**Write a C++ Program to generate first n terms of Fibonacci sequence.**
4.	**Write a program for recursive function.**
5.	**Write a C++ program to find both the largest and smallest number in a list of integers.**
6.	**Write a C++ Program to take two values from the user side and perform all kinds of Arithmetic Operation.**
7.	**Write a C++ program to sort a list of numbers in ascending order.**
8.	**Write a program to check Given Number is Even or Odd Using if-else in C++.**
9.	**Write a Program that shows the use of call by reference and Call by Value in C++.**
10.	**Write a Program to Demonstrate Friend Function and Friend Class.**
11.	**Write a Program to show the use of Inline Function.**
12.	**Create a Class named as “STUDENT”, Create two data members of that class:sid, smarks and Take two member Function of that class i.e.:GetData() to get the value of data member and ShowData() to Display the value of data members and call that function from the main function.**
13.	**Create a Class named as “EMPLOYEE”, Declare Three Data Members i.e.: eid,ename and e-salary and create two member functions of that class i.e.: GetDetails() to get the input of all the datamembers and PrintDetails() to Display all the details of the Employee. (note: create eid of int datatype, e-name is of char array data type and e-salary is of float data type.**
14.	**Write a Program to show the use of Function Overloading.**
15.	**Write a Program to show the use of Single level inheritance.**
16.	**Write a Program to show the use of multilevel inheritance.**
17.	**Write a Program to show the use of multiple inheritance.**
18.	**Write a Program to show the use of Hybrid inheritance.**
19.	**Write a Program to show the use of Virtual Base Class.**
20.	**Write a Program to show the use of a Default Constructor.**
21.	**Write a Program to show the use of a Parameterized Constructor.**
22.	**Write a Program to show the use of a Copy Constructor.**
23.	**Write a Program to show the use of ‘this’ pointer.**
24.	**Develop a program using pointers to derived classes.**